Simulation Modeling and Performance Evaluation

Lecture 5

Inventory System Analysis

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Inventory System (IS): Performance Analysis

- Problem Statement
  - Single product selling company
  - Demands come from customer
    - Satisfied immediately, if available
    - Periodically evaluated to place order
    - Arrival takes variable time
    - System is studied for a predefined time to
      - Minimize the average inventory cost

IS: Objectives and Goals

- Study different inventory policies
- To know the optimum inventory level
  - Order placement threshold (S_L)
  - Maximum inventory capacity (S_M)
    - $S_L =$ 20 20 20 20 40 40 40 60 60
    - $S_M =$ 40 60 80 100 60 80 100 80 100
- To know the optimum average inventory cost
  - Average ordering cost
  - Average holding cost
  - Average shortage cost

Conceptual Model

IS: System Description (1/2)

- System manages a single item
- Demands of customers
  - Arrive at random interval
  - Demand amount is variable
  - If demanded items are available
    - Satisfied immediate
  - Otherwise, keeps backlog
- Inventory Evaluation
  - Periodic with fixed interval
  - Evaluation determines
    - Need of an order placement
      - available items < threshold
    - Amount of order
      - # of items should not exceed the capacity
      - Minimization of unused capacity

IS: System Description (2/2)

- Order placement
  - Triggered by an evaluation
  - Determine ordering cost
- Order Arrival
  - Takes a random time after order placement
  - Backlog is clear first
  - Remaining items are added to the inventory
    - Non-zero available items iif zero backlog
- Period of study
  - Predefined number of months
  - No evaluation and order placement
    - Termination should occur before evaluation, if both have the same time occurrence

IS: State Variable(s)

- Inventory level, $I(t)$: # of items available at time t
- Initialization: initialized to a non-zero value
  - May be to the maximum inventory capacity
- It decreases, when a demand arrives
  - By the demand amount
- It increases, when a supply arrives
  - By the supply amount
- Its value might be negative
- The inventory level is evaluated at the beginning of each month
  - If $I(t) < S_L$, an order is placed,
    - S_L is order placement threshold

IS: Set of Events (1/3)

- Evaluation of the Inventory level: **Evaluation Event (e)**
  - Does not directly affect system state
    - Evaluates whether an order is to be placed
    - Determine the order amount
  - Thus, triggers the order placement
  - Occurs after a fixed interval
    - Beginning at each month

IS: Set of Events (2/3)

- Arrival of items from suppliers: **supply event (s)**
  - Clear the backlog, if any
  - Increases the inventory level by the supply amount
  - Supply arrival takes random time after an order is placed

- Arrival of a demand from customer: **demand event (d)**
  - Decreases the inventory level by the demand amount
  - Demand interval time is random
  - A demand might make the inventory level negative
    - If the demand amount is greater than $l(t)$

IS: Set of Events (3/3)

- Reaching the termination time: **termination event (t)**
  - Ends the simulation
    - Occurs only once in a simulation run
  - Forces the system to stop immediately
    - Cancel all events from the event list
  - Careful selection is necessary to allow this event to occur before event e
    - No need to evaluate the system and place an order if it ends immediately

IS: Input and Output Variables (1/2)

- Input variables:
  - Evaluation Interval
    - Fixed (beginning of each month)
  - Demand arrival interval
    - Exponential (mean 0.1 month)
  - Demand amount
    - 1, 2, 3 and 4 with probability 1/6, 1/3, 1/3 and 1/6, respectively
  - Supply arrival interval
    - Uniform within 0.5 to 1.0 month
  - Simulation termination time (120 months)
  - Maximum inventory level
  - Order placement threshold

IS: Input and Output Variables (2/2)

- Output Variables
  - Average inventory cost
  - Job-average output variable
    - Average ordering cost
      - Ratio of *total ordering cost* to # *of total evaluation*
  - Time-average output variable
    - Average holding cost
      - Holding area divided by evaluation period
    - Average shortage cost
      - Shortage area divided by evaluation period

Specification Model

IS: State Equation

- State variable is $l(t)$

$$l(t^{+}) = \begin{cases} I(t) + Z, & \text{if supply arrives} \\ I(t) - D, & \text{if demand arrives} \\ I(t), & \text{otherwise} \end{cases}$$

- Where, Z represents the amount of supply, and
- D represnets the amount of demand

- State Space:
  - $X_I = \{\cdots, -2, -1, 0, 1, 2, \cdots\}$

- Feasible Event set
  - $F_{I(t)} = \{e, d, s, t\} \; \forall I(t) \in X_I$

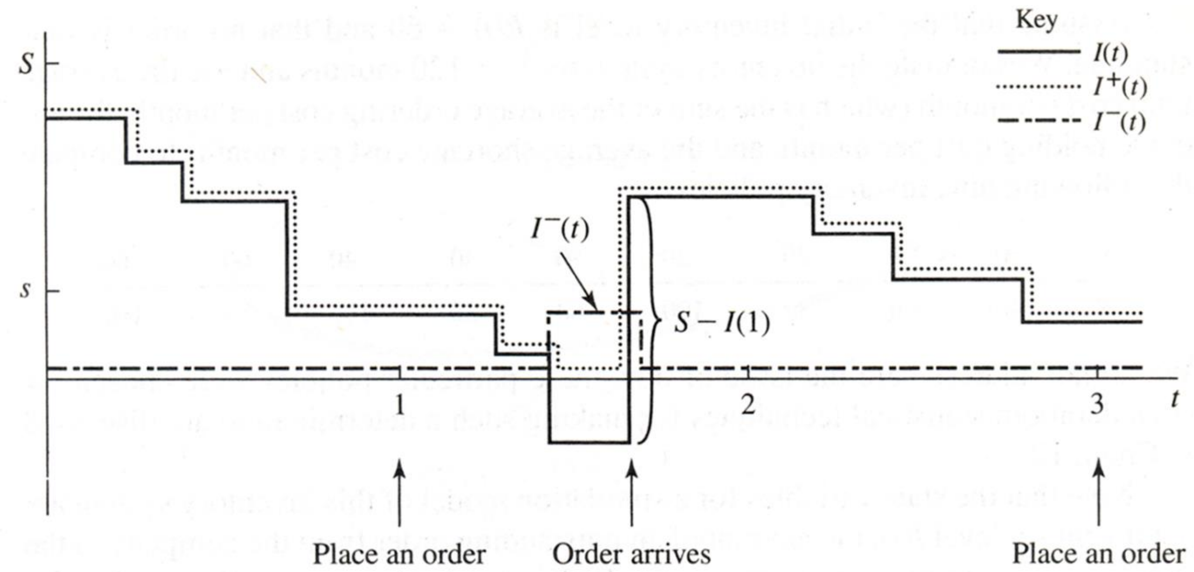
IS: Output Equations (1/4)

- Average ordering cost
  - Ordering cost per month
  - Ratio of total ordering cost to the number of months
  - Let c and C_o denote the ordering cost and total ordering cost, respectively
  - $c = I(Z) \times K + PZ$
  - Where K is set up cost, P is cost per item, and Z is an indicator function given by
  - $I(Z) = \begin{cases} 1, & I(t) > S_L \\ 0, & \text{otherwise} \end{cases}$
  - Every time an evaluation is done, total output cost is updated by, $C_o = C_O + c$
  - After the end of simulation, average ordering cost is found
    - $\overline{C_o} = \frac{C_O}{n}$, where n is number of evaluation months

IS: Output Equations (2/4)

- Average holding/shortage cost: time-average
  - Time average quantities are updated by every event
    - Because each event advances the simulated time
  - Holding/shortage cost depends on # of items
    - Holding cost is associated with $l(t) > 0$
    - Shortage cost is associated with $l(t) < 0$
    - However, both costs are positive
  - Two additional variables are defined
    - $I^{+}(t) = \max\ (I(t), 0)$
    - $I^{-}(t) = \max\ (-I(t), 0)$

IS: Output Equations (3/4)



- Time-average # of items in system

$$\overline{I^+} = \frac{\int_0^n I^+(t)dt}{n}$$

- Time-average # of items in backlog

$$\overline{I^-} = \frac{\int_0^n I^-(t)dt}{n}$$

IS: Output Equations (4/4)

- Average holding cost/month $= h\overline{I^+}$
  - Where, h is the holding cost per item per month
- Average holding cost/month $= \pi\overline{I^-}$
  - Where, π is the backlog cost per item per month

IS: Input Data Modeling (1/2)

- Evaluation Interval: deterministic
  - At the beginning of each month
- Demand Interval: Random
  - IID Exponential random variable
  - Mean is 0.1 month
- Demand Amount: Random
  - IID random variable with the PMF
  - $P_D(d) = \begin{cases} \frac{1}{6}, & d = 1, 4 \\ \frac{1}{3}, & d = 2, 3 \\ 0, & \text{otherwise} \end{cases}$

IS: Input Data Modeling (1/2)

- Delivery Lag: Random
  - IID uniform random variable with parameters
    - $a = 0.5$ month
    - $b = 1.0$ month
- Setup cost (K): fixed - $K = \$32$
- Item cost (P): fixed - $P = \$3$
- Item holding cost (h): fixed - $h = \$1$
- Item backlog cost (π): fixed - $\pi = \$5$

Computational Model

Computational Model: Simulator

Member variables

- Clock_
- eventList_

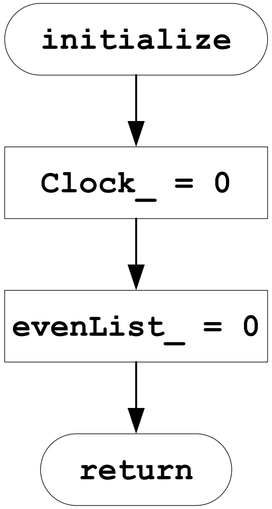
Member Functions

- Initialize ()
- Run ()
- addEvent (Event *e)
- Event* removeEvent ()

Computational Model: Simulator

initialize () function

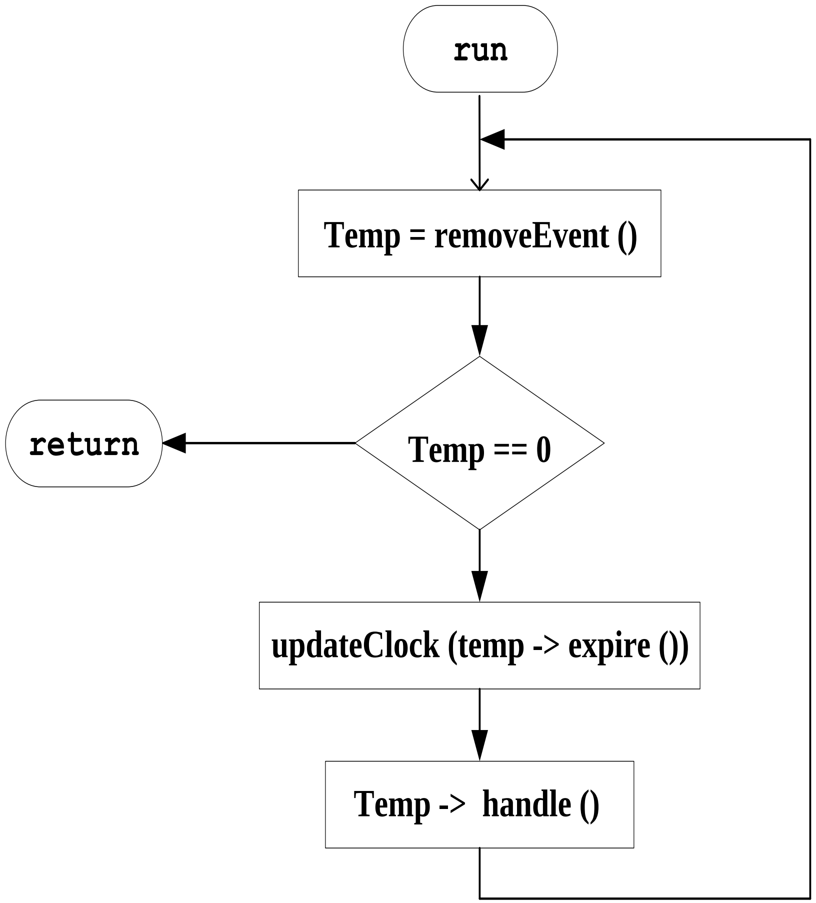
- Initialize the simulator
- Set the clock_ to 0
- Set the eventList_ to null



Computational Model: Simulator

run () function

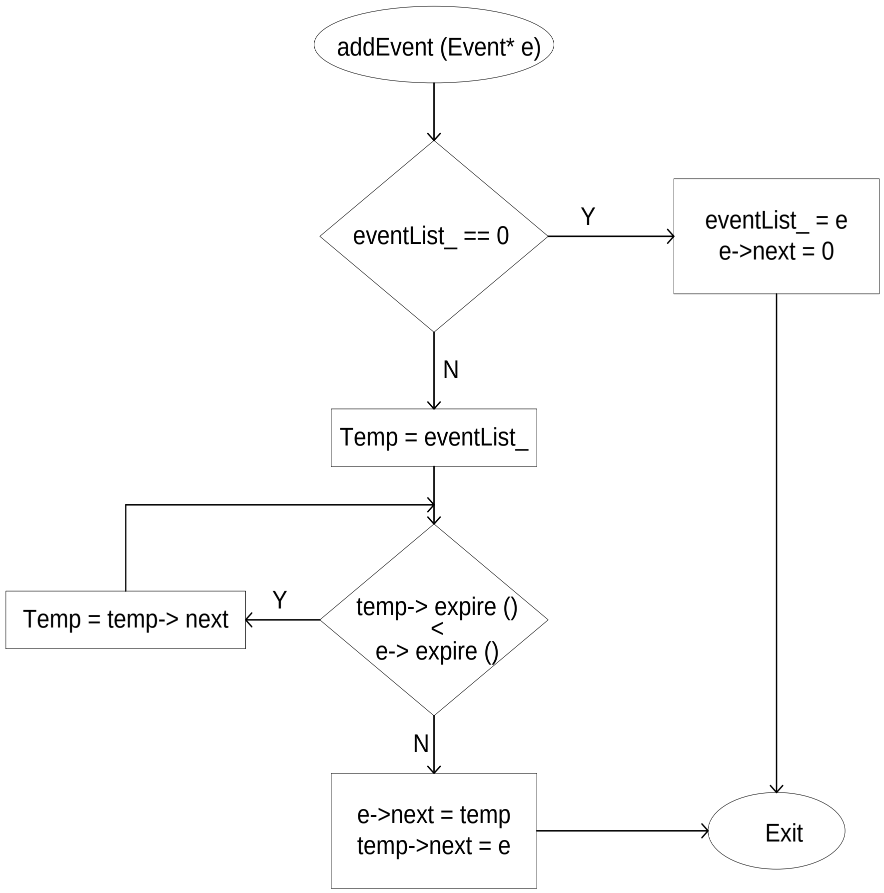
- Schedule the events
  - First event in the eventList_ is scheduled
  - If there is no event in eventList_, simulation is terminated
- Update the simulation time clock_ to the current time
- Call the event handling function
- Move to the next event in the list



Computational Model: Simulator

addEvent () function

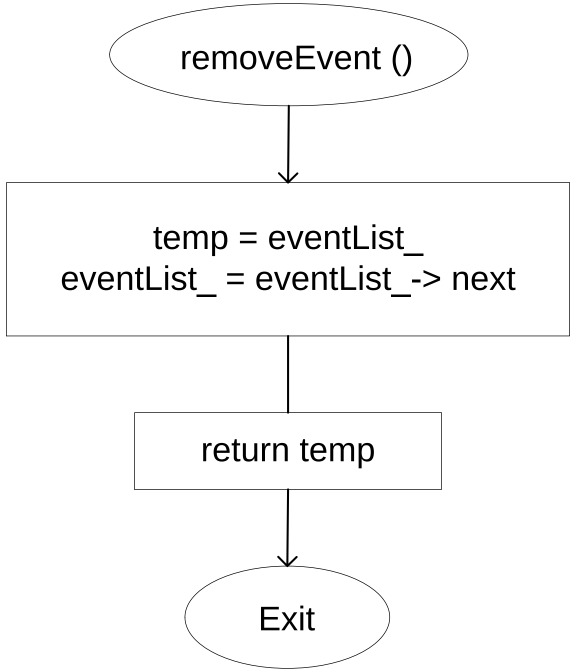
- Add an event in the eventList_
- Check whether the event is added
  - in an empty list
  - as first element in the list
  - in the middle of the list
  - as last element of the list
- Flow chart shows two cases
  - Empty list
  - Middle of a list



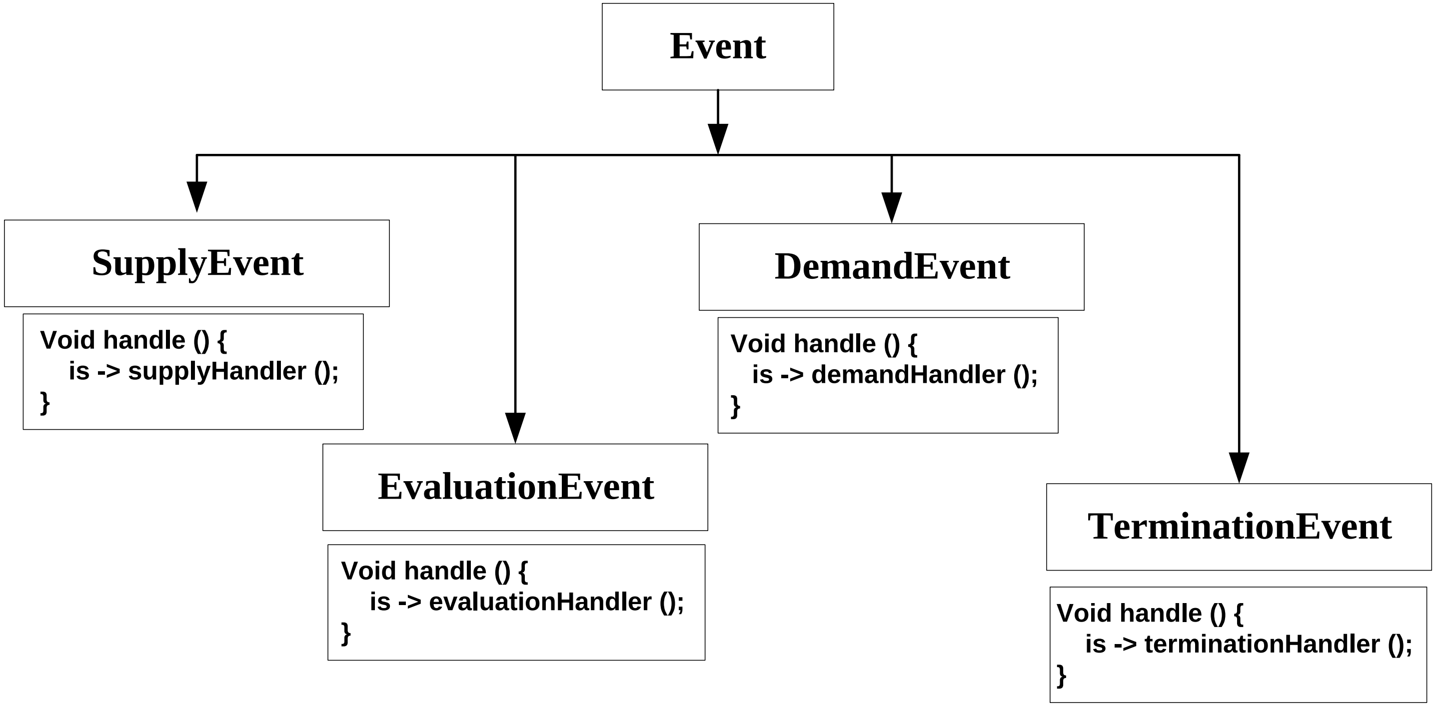
Computational Model: Simulator

removeEvent () function

- Remove the first event of the evenList_
  - Return the event
- If the list is empty returns null



Computational Model: Events



Computational Model: Inventory

Member variables

- State variables
  - ivLevel_
- Events
  - d_, e_, s_, t_
- Constants
  - maxLevel_, minLevel_, n
  - minLag_, maxLag_
  - setupCost_, itemCost_
  - arrivalMin_, demandMin_
- Statistical Variables
  - areaHolding_, areaShortage_
  - orderingCost_
- Temporary Counters
  - orderAmount_
  - timeLastEvent_
  - evalId_, arid_, demndId_

Member Functions

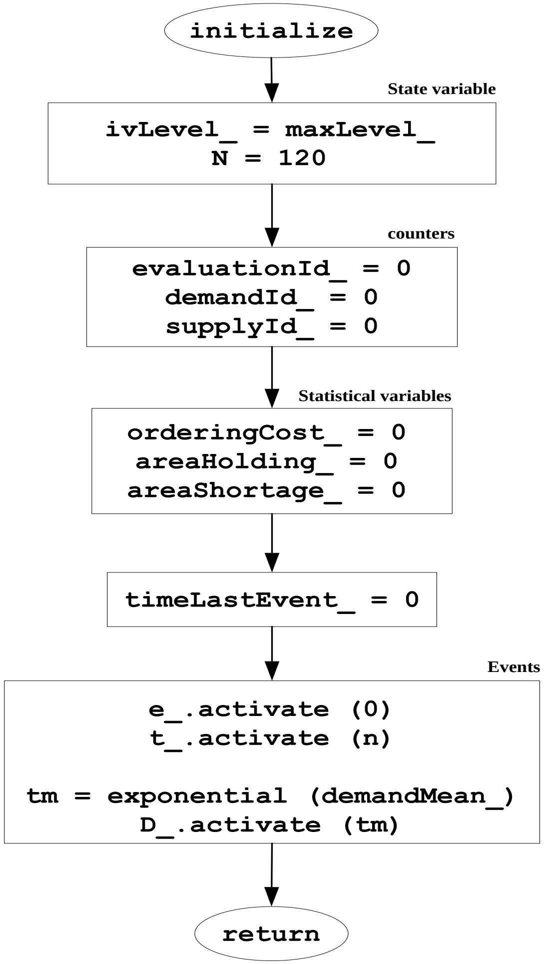
- Initialize ()
- demandHandler()
- evaluationHandler()
- supplyHandler()
- terminationHandler()

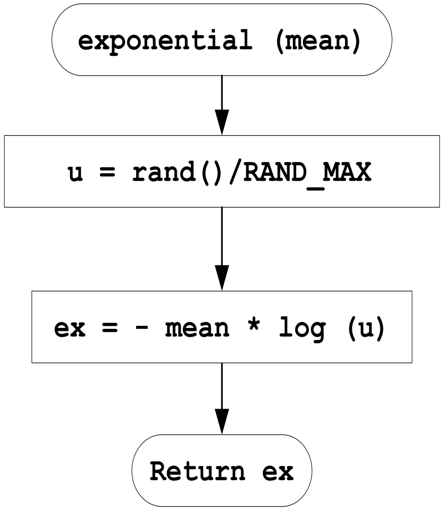
- Exponential (mean)
- uniformRand(min, max)
- discreteRand()

Computational Model: Inventory

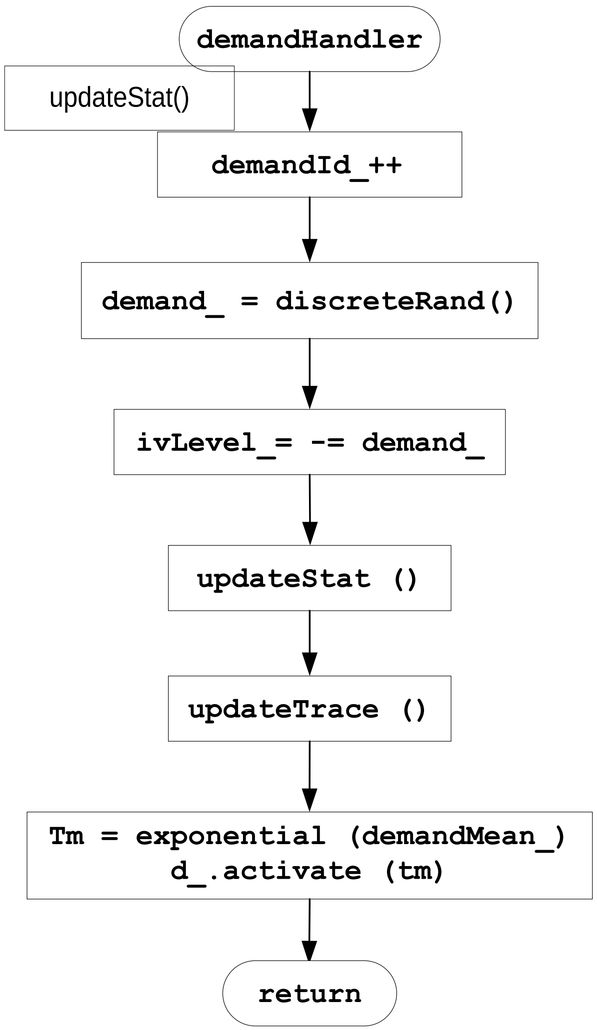
<u>initialize () function</u>

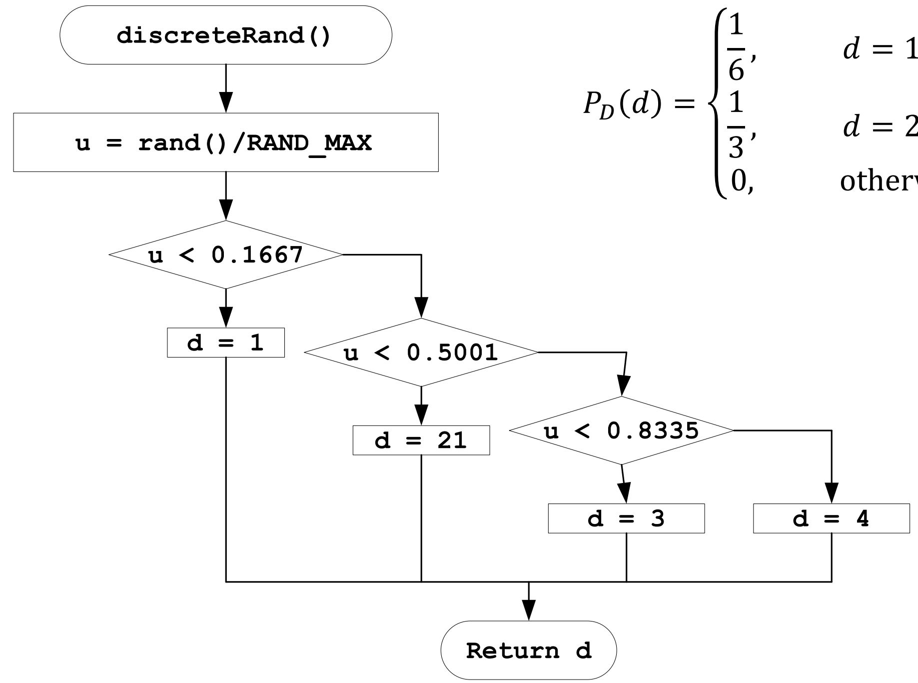
- Initialize the state variables
  - ivLevel_ = 0
- Initialize input variables
  - n = 120
  - demandMean_ = 0.1
  - $S_L = 20, S_M = 40$
  - minLag_ = 0.5, maxLag_ = 1.0
- Initialize counter and output variables
- Finally, activate evaluation and demand events





IS: Demand Handler





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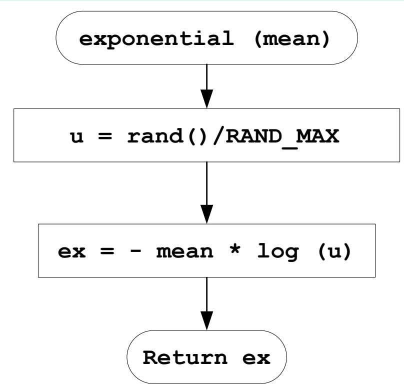
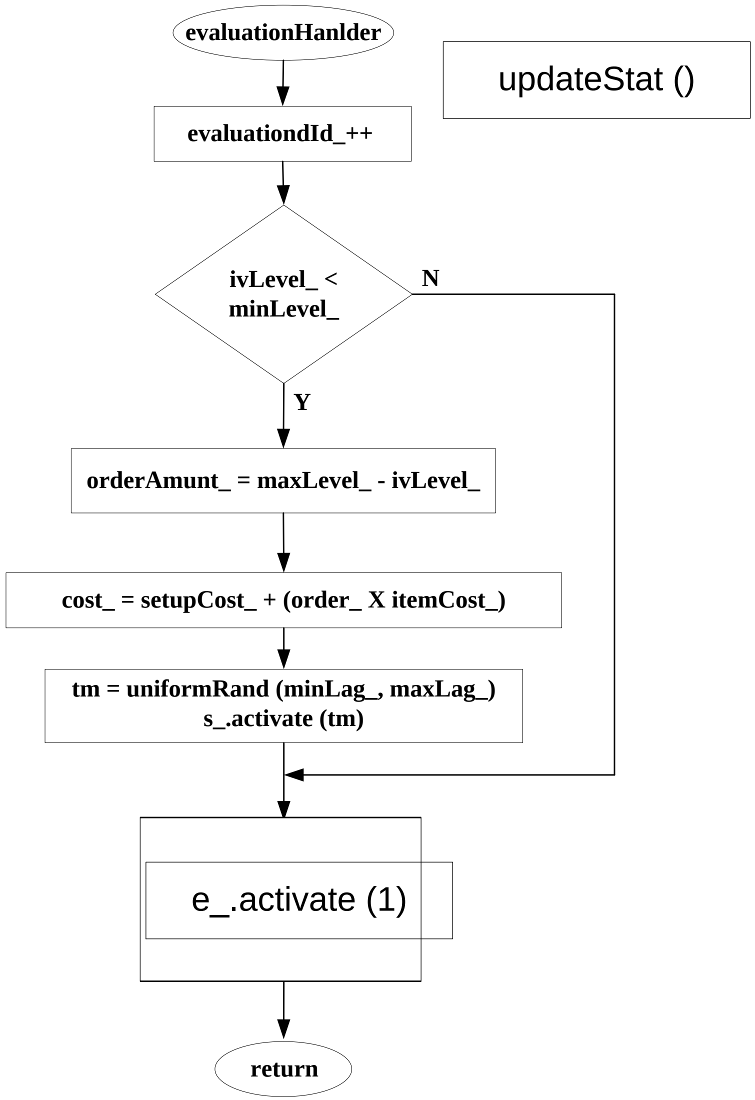
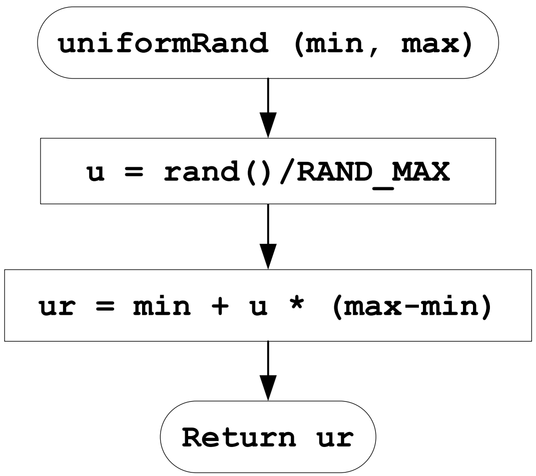


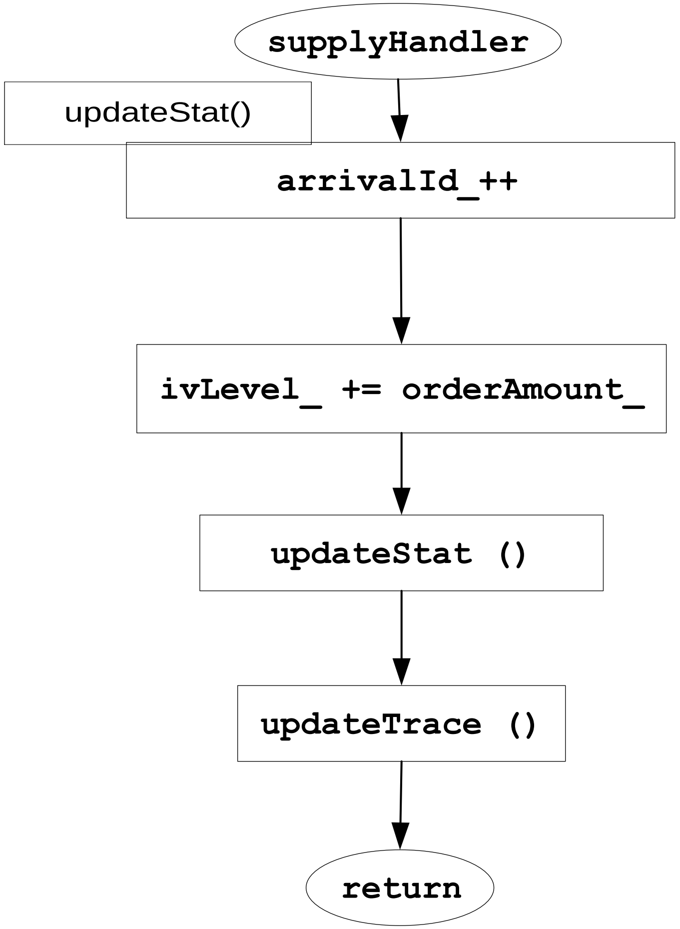
IS: Evaluation Handler





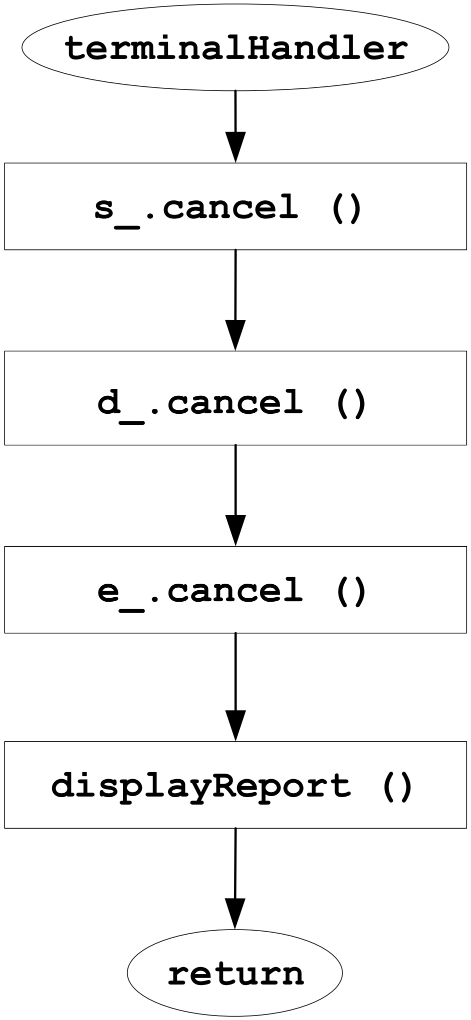
IS: Supply Handler

- Increase the ivLevel_ by the orderAmount_
- Does not schedule a supply event
  - supplyEvent is schedule by the evaluation event



IS: Termination Handler

- Remove all the event from the evenList_
- Next iteration of the run function will remove a null from the evenList_
- Terminate the simulation



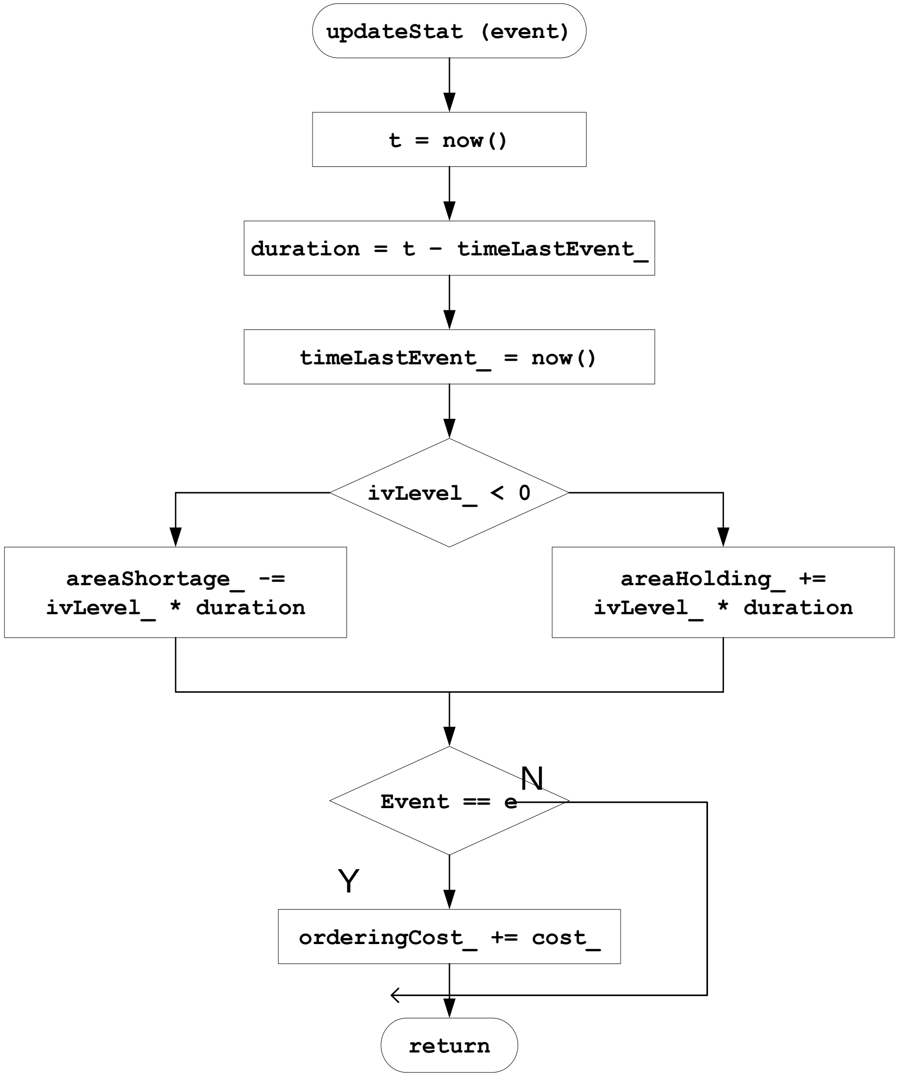
IS: Update Statistical Variables

avgOrderingCost = orderingCost_ / 120.0

avgHoldingCost = holdingCost_ * areaHolding_ / 120.0

avgShortageCost = shortageCost_ * areaShortage_ / 120.0;

avgCost = avgOrderingCost + avgShortageCost + avgHoldingCost



Different Policies

- $S_L = 20 \quad 20 \quad 20 \quad 20 \quad 40 \quad 40 \quad 40 \quad 60 \quad 60$
- $S_M = 40 \quad 60 \quad 80 \quad 100 \quad 60 \quad 80 \quad 100 \quad 80 \quad 100$